Bureau
International des
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Mesures

Le système international d'unités
The International System of Units

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Annexe 2
Annex 2

Valeurs recommandées des fréquences étalons destinées à la mise en pratique de la définition du mètre et aux représentations secondaires de la seconde

Recommended values of standard frequencies for applications including the practical realization of the metre and secondary representations of the second

Partie 2
Part 2

Hydrogen Atom ()

Hydrogen Atom ()



CCL-CCTF Frequency Standards Working Group

Bureau International
des Poids et Mesures

Recommended values of standard frequencies for applications including the practical realization of the metre and secondary representations of the second
Part 2: Hydrogen Atom ()

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Appendix 2

1. Recommended value CIPM REC 2 (2015, E) of the frequency

$f\left({}^{1}\mathrm{H}\right) = 1\ 233\ 030\ 706\ 593\ 514\ \mathrm{Hz}$

equivalent to

$\lambda\left({}^{1}\mathrm{H}\right) = 243\ 134\ 624.626\ 044\ \mathrm{fm},$

with an estimated relative standard uncertainty of 9×10^{-15}.

The $1S-2S$ transition is excited by two photons of the frequency 1233 THz

2. Source data

Adopted value $f\left({}^{1}\mathrm{H}\right) = 1\ 233\ 030\ 706\ 593\ 514\ \mathrm{Hz}$

$u_c/y = 9 \times 10^{-15}$

calculated from

$2 \times f\left({}^{1}\mathrm{H}\right)$/ Hz	u/ Hz	source data
2 466 061 413 187 035	10	[2]
2 466 061 413 187 018	11	[3]

as a weighted mean of the above values divided by two.

Given the noticeable drift of the values of the past measurements [3] and the fact that the values come from a single laboratory the CCTF considered it prudent to attribute an increased standard uncertainty by a factor of three.

^{1}H neutral atom, 1S − 2S unperturbed optical transition

References

[1] CIPM REC 2 (2015, E) BIPM (2015) *Updates to the list of standard frequencies*. (Paris: International Bureau of Weights and Measures, Bureau international des poids et mesures) .

[2] C. G. Parthey, A. Matveev, J. Alnis, B. Bernhardt, A. Beyer, R. Holzwarth, A. Maistrou, R. Pohl, K. Predehl, T. Udem, T. Wilken, N. Kolachevsky, M. Abgrall, D. Rovera, C. Salomon, P. Laurent, and T. W. Hänsch: Improved Measurement of the Hydrogen 1S—2S Transition Frequency, *Phys. Rev. Lett.* **107**, 203001 (2011).

[3] A. Matveev, Ch. G. Parthey, K. Predehl, J. Alnis, A. Beyer, R. Holzwarth, Th. Udem, T. Wilken, N. Kolachevsky, M. Abgrall, D. Rovera, Ch. Salomon, Ph. Laurent, G. Grosche, O. Terra, Th. Legero, H. Schnatz, S. Weyers, B. Altschul, Th. W. Hänsch: Precision Measurement of the Hydrogen 1S − 2S Frequency via a 920 − km Fiber Link, *Phys. Rev. Lett.* **110**, 230801 (2013).